The Bayesian Infinitesimal Jackknife for Variance

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Abstract

The frequentist variability of Bayesian posterior expectations can provide meaningful measures of uncertainty even when models are misspecified. However, standard methods to approximate the frequentist covariance of posterior expectations — namely, those using the MAP estimator or the nonparametric bootstrap — can be practically inconvenient. In particular, the MAP estimator may behave poorly in moderate dimensions, and the bootstrap requires computing the posterior for many different bootstrap datasets. We develop another method, the infinitesimal jackknife (IJ), for estimating the frequentist variability of posterior expectations. The IJ is a method for computing asymptotic frequentist covariances of smooth functionals of exchangeable data using the "influence function" of robust statistics. We show that the influence function for posterior expectations is given by a simple posterior covariance, and that the IJ covariance estimate is easily computed from a single set of posterior samples. Under conditions similar to those required for a Bayesian central limit theorem, we prove that the IJ covariance estimate is asymptotically equivalent to that given by the sandwich covariance and the bootstrap, but avoids reliance on an accurate MAP estimate or on bootstrap samples. We demonstrate the accuracy and computational benefits of the IJ covariance estimates with simulated and real-world experiments.

Keywords: Influence function, sandwich covariance, infinitesimal jackknife, Bernstein-von Mises theorem, Bayesian robustness

1 Introduction

Suppose we want to use Bayesian methods to infer some parameter or quantity of interest from exchangeable data. The Bayesian posterior is often summarized by a point estimate and measure of subjective uncertainty, which we will respectively take to be the posterior mean and posterior variance. The computed posterior mean is a function of the observed data, and so has a well-defined frequentist variance under re-sampling of the exchangeable data. The Bayesian posterior variance of a quantity and the frequentist variance of its mean are conceptually distinct: the posterior variance can be thought of as describing the uncertainty of subjective beliefs under the assumption of correct model specification, and the frequentist variance captures the sensitivity of the Bayesian point estimate to re-sampling of the observed data. In general, both may be of interest. Even a strict Bayesian can meaningfully interpret the frequentist variance of the posterior mean as a *sensitivity analysis*, quantifying how much the Bayesian point estimate would vary had we received different but realistic data. Such a sensitivity analysis is particularly germane when the data is known to have come from a random sample from a real population.

In general, the frequentist and Bayesian variances are not only conceptually distinct but can be numerically very different. It is true that, with independent and identically distributed (IID) data, some reasonable regularity conditions, a correctly specified model, and a parameter vector of fixed dimension, the frequentist and Bayesian variances coincide asymptotically as a consequence of the Bernstein–von Mises theorem, a.k.a. the Bayesian central limit theorem. However, under *model misspecification*, the equality between the frequentist and Bayesian variances no longer holds in general, even asymptotically (Kleijn and van der Vaart, 2012) — and most models are misspecified.

When posterior means are estimated using Markov Chain Monte Carlo (MCMC), the two standard tools for estimating the frequentist variability of Bayesian posterior expectations —

the nonparametric bootstrap (henceforth simply "bootstrap") and the Laplace approximation — can be each be problematic, though for different reasons. To compute a bootstrap estimate of the frequentist variance of a posterior mean, one repeatedly draws "bootstrap datasets," with replacement from the original set of datapoints. For each bootstrap dataset, one then computes the posterior mean for each set of re-sampled data (Huggins and Miller, 2023). Though straightforward to implement and naively parallelizable, the bootstrap remains computationally expensive, since a separate posterior estimate (e.g., a run of MCMC) must be computed for each bootstrap dataset. Alternatively, it is straightforward to estimate frequentist variance of a Laplace approximation to the posterior. Specifically, one can do so by computing the "sandwich covariance" estimate of the maximum *a posteriori* (MAP) estimate at which the Laplace approximation is centered (Stefanski and Boos, 2002). This approach essentially uses the standard consistent estimator of the limiting covariance derived by Kleijn and van der Vaart (2012). However, this approach requires using a particular Laplace approximation to the posterior rather than MCMC and, as we show below, the Laplace approximation can be practically and conceptually problematic. For example, unlike true posterior draws, the Laplace approximation is not invariant to reparameterization, and, for some parameterizations the Laplace approximation can fail catastrophically, especially in hierarchical models.

In the present paper, we provide an estimator of the frequentist variability of posterior expectations that enjoys the model-agnostic properties of the bootstrap, but can be computed from a single run of MCMC. Our method is based on the *infinitesimal jackknife* (IJ) variance estimator, a classical frequentist variance estimator derived from the *empirical influence function* of a statistical functional. We show that the Bayesian empirical influence function is easily computable as a particular set of posterior covariances, the variance of which is a consistent estimator of the frequentist variance of a posterior expectation when a Bayesian central limit theorem (BCLT) holds for all model parameters.

Finally, though our present concern is frequentist variance estimates, the empirical influence function is a powerful statistical tool, with applications in robustness, cross validation, and bias estimation. Consequently, our results for the influence function of Bayesian posteriors and a BCLT for expectations of data-dependent functions are of broader interest. We leave off exploring these topics for future work.

2 Setup and Motivation

Suppose we observe N data points, $\mathcal{X} = (x_1, \ldots x_N)$. We will assume that the entries of $\mathcal{X}$ are independently and identically distributed (IID) according to a distribution $\mathbb{F}$. In a slight abuse of notation, we will write $\mathbb{F}(\mathcal{X})$ for $\prod_{n=1}^{N} \mathbb{F}(x_n)$. Throughout, the distribution $\mathbb{F}$ will be taken as fixed but unknown. A Bayesian typically models the distribution of $\mathcal{X}$ by an IID parametric model given on some parameters $\theta \in \Omega_\theta \subseteq \mathbb{R}^{D_\theta}$:

$$\mathbb{P}(\mathcal{X}|\theta) = \prod_{n=1}^{N} \mathbb{P}(x_n|\theta) = \exp\left(\sum_{n=1}^{N} \ell(x_n|\theta)\right) \quad \text{where} \quad \ell(x_n|\theta) := \log \mathbb{P}(x_n|\theta).$$

In the present work, we admit that the model may be misspecified, i.e., there may be no $\theta \in \Omega_\theta$ such that $\mathbb{P}(x_n|\theta)$ is the same distribution as $\mathbb{F}$.

We will additionally posit a prior over the parameters θ, and assume this prior can be expressed as a density $\pi(\cdot)$ with respect to the Lebesgue measure on $\mathbb{R}^{D_\theta}$. Then we can compute the posterior expectation of a vector-valued quantity $g(\theta) \in \mathbb{R}^{D_g}$ using Bayes' rule:

$$\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[g(\theta)\right] = \frac{\int_{\Omega_\theta} g(\theta) \exp\left(N \frac{1}{N}\sum_{n=1}^{N} \ell(x_n|\theta)\right) \pi(\theta) d\theta}{\int_{\Omega_\theta} \exp\left(N \frac{1}{N}\sum_{n=1}^{N} \ell(x_n|\theta)\right) \pi(\theta) d\theta}. \tag{1}$$

The quantity $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[g(\theta)\right]$ is a function of the data $\mathcal{X}$, through its dependence on the random function $\theta \mapsto \frac{1}{N}\sum_{n=1}^{N} \ell(x_n|\theta)$. We will be primarily concerned with the problem of how to estimate the variability of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ under sampling from $\mathbb{F}(\mathcal{X})$, when N is large, using Markov Chain Monte Carlo (MCMC) samples from the posterior $\mathbb{P}(\theta|\mathcal{X})$, in the presence of possible misspecification. Specifically, let $\xrightarrow[N\to\infty]{prob}$ and $\xrightarrow[N\to\infty]{dist}$ respectively denote convergence in probability and distribution under sampling from $\mathbb{F}(\mathcal{X})$. We will articulate conditions under which

$$\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)] \xrightarrow[N\to\infty]{prob} \mu^g < \infty \quad \text{and} \quad \sqrt{N}\left(\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)] - \mu^g\right) \xrightarrow[N\to\infty]{dist} \mathcal{N}\left(\cdot|0, V^g\right) \tag{2}$$

for some constants μ^g and V^g. Under these conditions, our contribution is to form a consistent estimator of V^g that can be computed from a single set of draws from $\mathbb{P}(\theta|\mathcal{X})$, and so approximated from a single run of MCMC. We emphasize that our objective of establishing Eq. 2 is weaker than estimating the limiting variance of $\sqrt{N}\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$, which may not even exist, but which would coincide with V^g under an additional assumption of uniform integrability (see, e.g., Theorem 10.3.6 of Dudley (2018)).

2.1 Notation

We will begin by defining some notation. Given a distribution $\mathbb{P}(t)$ and an integrable function $\phi(t)$, we will use the expectation operator as a formal shorthand for the integral $\underset{\mathbb{P}(t)}{\mathbb{E}}[\phi(t)] = \int \phi(t)\mathbb{P}(dt)$. Any quantities not explicitly part of the distribution beneath the expectation operator should be thought of as fixed. Covariances and variances are treated similarly, i.e. $\underset{\mathbb{P}(t)}{\mathrm{Cov}}(\phi(t), \psi(t)) = \underset{\mathbb{P}(t)}{\mathbb{E}}\left[\phi(t)\psi(t)^T\right] - \underset{\mathbb{P}(t)}{\mathbb{E}}[\phi(t)]\underset{\mathbb{P}(t)}{\mathbb{E}}[\psi(t)]^T$. We will denote sample covariances over an index $n \in \{1, \ldots, N\} = [N]$ as $\underset{n\in[N]}{\widehat{\mathrm{Cov}}}(\cdot, \cdot)$.

We denote partial derivatives with respect to θ with bracketed subscripts. For example, for a function $\psi(\theta)$ taking values in $\mathbb{R}^{D_\psi}$, let $\psi_{(k)}(\theta)$ denote the array of k-th order partial derivatives with respect to θ, i.e., $\psi_{(k)}(\theta) := \left.\frac{\partial^k \psi(\theta)}{\partial \theta^k}\right|_\theta$. If ψ depends on any variables other than θ, those variables will be taken as fixed in the partial differentiation. In general, $\psi_{(k)}(\theta)$

is an array with $D_\psi D_\theta^k$ entries, and we will introduce special notation as needed when summing over the array's entries. But we will mostly be able to avoid such special notation by interpreting the following special cases in standard matrix notation: when $D_\psi = 1$ and $k = 1$, $\psi_{(1)}(\theta)$ is taken to be a D_θ-column vector; when $D_\psi > 1$ and $k = 1$, $\psi_{(1)}(\theta)$ is taken to be a $D_\psi \times D_\theta$ matrix; when $D_\psi = 1$ and $k = 2$, $\psi_{(2)}(\theta)$ is taken to be a $D_\theta \times D_\theta$ matrix; and when $k = 0$, $\psi_{(0)}(\theta)$ is just $\psi(\theta)$.

Our proposal and existing methods for estimating V^g will depend on some key quantities to which we give symbols in Definition 1. In the left hand column of Definition 1 we have finite-sample quantities, and on the right hand side we have the corresponding population quantities. In Section 3 below we will state precise conditions under which all the quantities of Definition 1 are well-defined. We will use x_* to denote a generic single datapoint, in contrast to specific observed datapoints, x_n.

Definition 1.

Finite-sample quantity	Population quantity
$\hat{\mathscr{L}}(\theta) := \frac{1}{N} \sum_{n=1}^{N} \ell(x_n \vert \theta) + \frac{1}{N} \log \pi(\theta)$	$\mathscr{L}(\theta) := \underset{\mathbb{F}(x_*)}{\mathbb{E}} \left[\ell(x_* \vert \theta) \right]$
$\hat{\theta} := \underset{\theta \in \Omega_\theta}{\mathrm{argmax}}\, \hat{\mathscr{L}}(\theta)$	$\theta_\infty := \underset{\theta \in \Omega_\theta}{\mathrm{argmax}}\, \mathscr{L}(\theta)$
$\hat{\Sigma} := \frac{1}{N} \sum_{n=1}^{N} \ell_{(1)}(x_n \vert \hat{\theta}) \ell_{(1)}(x_n \vert \hat{\theta})^T$	$\Sigma := \underset{\mathbb{F}(x_*)}{\mathrm{Cov}} \left(\ell_{(1)}(x_* \vert \theta_\infty) \right)$
$\hat{\mathcal{I}} := -\hat{\mathscr{L}}_{(2)}(\hat{\theta})$	$\mathcal{I} := -\underset{\mathbb{F}(x_*)}{\mathbb{E}} \left[\ell_{(2)}(x_* \vert \theta_\infty) \right]$ ◀

We will take the population quantities in the right column of Definition 1 to be fixed (though, in general, unknown) and study the sampling behavior of quantities in the left column. That is, we will study the asymptotic behavior of formal Bayesian posterior quantities as a function of a fixed data generating process, following much of the literature on the frequentist behavior of Bayesian estimators (e.g., van der Vaart (2000)). Our notation

Algorithm 1 Sandwich covariance

Requires: Differentiable $\theta \mapsto \hat{\mathscr{L}}(\theta)$

$\hat{\theta} \leftarrow \mathrm{argmax}_{\theta \in \Omega_\theta} \hat{\mathscr{L}}(\theta)$

$\hat{\Sigma}_\theta \leftarrow \hat{\mathcal{I}}^{-1} \hat{\Sigma} \hat{\mathcal{I}}^{-1}$

return $\hat{V}^{\mathrm{Sand}} \leftarrow g_{(1)}(\hat{\theta}) \hat{\Sigma}_\theta g_{(1)}(\hat{\theta})^\intercal$

Algorithm 2 Bootstrap covariance

Requires: Posterior sampler $\mathcal{X}^* \mapsto \theta^1, \ldots, \theta^M \sim \mathbb{P}(\theta|\mathcal{X}^*)$ (e.g. MCMC)

for $b = 1, \ldots, B$ **do**

$\mathcal{X}^* \leftarrow$ Sample with replacement from the rows of $\mathcal{X}$

Sample $\theta^1, \ldots, \theta^M \sim \mathbb{P}(\theta|\mathcal{X}^*)$

$\mathscr{G}^b \leftarrow \frac{1}{M} \sum_{m=1}^M g(\theta^m)$

end for

return $\hat{V}^{\mathrm{Boot}} \leftarrow \underset{b \in [B]}{\widehat{\mathrm{Cov}}} \left(\sqrt{N} \mathscr{G}^b \right)$

Algorithm 3 IJ covariance (our proposal)

Requires: Log likelihood $\theta, x_n \mapsto \ell(x_n|\theta)$, posterior samples $\theta^1, \ldots, \theta^M \sim \mathbb{P}(\theta|\mathcal{X})$

for $n = 1, \ldots, N$ **do**

$\hat{\psi}_n \leftarrow N \underset{m \in [M]}{\widehat{\mathrm{Cov}}} \left(\ell(x_n|\theta^m), g(\theta^m) \right)$

end for

return $\hat{V}^{\mathrm{IJ}} \leftarrow \underset{n \in [N]}{\widehat{\mathrm{Cov}}} \left(\hat{\psi}_n \right)$

Figure 1: Three algorithms for estimating V^g.

for stochastic convergence reflects these assumptions. In particular, order notation $O(\cdot)$ applies as $N \to \infty$, and probabilistic order notation $O_p(\cdot)$ denotes order in probability over $\mathbb{F}(x_*)$. Tilde order notation $\tilde{O}_p(\cdot)$ neglects factors that are sub-polynomial in N.

Let $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ denote the usual vector norms. When applied to an array of derivatives, the norms refer to the vector norm applied to the vectorized version of the array.

2.2 Three methods for estimating V^g

There are two well-known classical methods for estimating V^g: the sandwich covariance and the bootstrap. The sandwich covariance requires computation of the maximum a posteriori (MAP) estimate, and is not available from MCMC samples. The bootstrap requires running many MCMC chains with slightly different data sets, which can be computationally expensive. We describe a third method, the infinitesimal jackknife (IJ) covariance estimate, which is also classical, but which has not been previously applied to MCMC estimates of posterior means. All three are succinctly summarized in Figure 1.

The sandwich covariance approach to estimating V^g is given in Algorithm 1. To compute the sandwich covariance estimate, one first forms the MAP estimate, $\hat{\theta}$. Then one estimates the sampling variability of $\hat{\theta}$ using an asymptotic approximation to the sampling distribution of M–estimators under misspecification. From the sampling variability of $\hat{\theta}$, one then uses the delta method to estimate the sampling variability of $g(\hat{\theta})$ (Stefanski and Boos (2002), Chapter 7 of van der Vaart (2000)).[1] When a sufficiently strong BCLT applies, it is known that $g(\hat{\theta})$ consistently estimates $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ and so the sandwich covariance estimate consistently estimates V^g (more details will follow in Section 3.1 below). However, the MAP approximation is not parameterization–invariant, and can behave pathologically in particular datasets, even when there exists a parameterization in which a BCLT applies.

When the MAP and sandwich covariance are unreliable, one typically turns to MCMC estimates. To estimate V^g from MCMC samples, a classical method is the bootstrap (Efron and Tibshirani, 1994; Huggins and Miller, 2023). The bootstrap is shown in Algorithm 2. The primary difficulty with the bootstrap is that it requires B different MCMC chains, which can be computationally expensive; often, in practice, even a single run of MCMC

[1] It would arguably be more standard to describe the covariance estimator for $\hat{\theta}$ as the "sandwich covariance" and the covariance estimator of $g(\hat{\theta})$ as the "delta method covariance." But the IJ covariance estimate is sometimes described as the "functional delta method" (van der Vaart (2000) Chapter 20), so the authors believe that our nomenclature is the least ambiguous option.

requires considerable computing resources. When the MCMC noise can be made arbitrarily small, the bootstrap is a consistent estimator of V^g in the fixed-dimension case under conditions given by Huggins and Miller (2023).

We propose an estimator of V^g that can be computed from a single MCMC run using only the tractable $\ell(x_n|z,\theta)$. We call our estimator the *infinitesimal jackknife (IJ) covariance estimator*, and denote it as V^{IJ}. The IJ covariance estimator is given by the sample variance of a set of simple posterior covariances, as given in Eq. 3.

$$\psi_n := N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}} \left(g(\theta), \ell(x_n|\theta)\right) \text{ (Influence score)} \qquad V^{\mathrm{IJ}} := \underset{n\in[N]}{\widehat{\mathrm{Cov}}} \left(\psi_n\right). \tag{3}$$

In practice, the posterior covariances of Eq. 3 typically need to be estimated with MCMC. We denote by $\hat{V}^{\mathrm{IJ}}$ the corresponding estimate of V^{IJ} (see Algorithm 3 for a precise statement). For any fixed N, the output $\hat{V}^{\mathrm{IJ}}$ of Algorithm 3 approaches V^{IJ} under a well-mixing MCMC chain as the number of MCMC samples $M \to \infty$. We will theoretically analyze V^{IJ}, though our experiments will all employ $\hat{V}^{\mathrm{IJ}}$. Importantly, Algorithm 3 requires only a single run of MCMC without relying on accuracy of the MAP, as required by the sandwich estimator, or running B different MCMC samplers, as required by the bootstrap estimator. Our key result, given in Theorem 2 below, states general conditions under which $V^{\mathrm{IJ}} \xrightarrow[N\to\infty]{prob} V^g$.

3 Theoretical results

Before we state our formal results in Section 3.1, we will present some high-level intuition to motivate the IJ covariance estimate, as well as our data-dependent Bayesian central limit theorem of Section 3.1.

For simplicity let us for the moment take the quantity of interest, $g(\theta)$, to simply be

$g(\theta) = \theta$. Following Kleijn and van der Vaart (2012), for example, we might informally expect the following central limit theorem results to hold:

$$
\begin{aligned}
N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta\right) =& \mathcal{I}^{-1} + O_p(N^{-1}) && \text{Bayesian CLT (BCLT)} \\
\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\theta\right] =& \hat{\theta} + O_p(N^{-1}). && \text{Bayesian CLT (BCLT)} \\
\sqrt{N}\left(\hat{\theta} - \theta_\infty\right) & \xrightarrow[N\to\infty]{dist} \mathcal{N}\left(\cdot|0, \mathcal{I}^{-1}\Sigma\mathcal{I}^{-1}\right). && \text{Frequentist CLT (FCLT)}
\end{aligned}
\tag{4}
$$

Here and throughout the paper, we somewhat loosely use the term "Bayesian CLT" (BCLT) to refer to results concerning the asymptotic properties of the posterior measure, and "Frequentist CLT" (FCLT) for results concerning properties under the data sampling measure. Both BCLT and FCLT results are typically proven using frequentist tools.

Given the work of Kleijn and van der Vaart (2012), we know that the limiting covariance we are trying to estimate is the sandwich covariance matrix, $V^g = \mathcal{I}^{-1}\Sigma\mathcal{I}^{-1}$. Since $N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta\right)$ converges to $\mathcal{I}^{-1}$, but $\mathcal{I}^{-1} \neq \mathcal{I}^{-1}\Sigma\mathcal{I}^{-1}$ in general unless the model is correctly specified, V^g cannot, in general, be accurately approximated by the Bayesian posterior covariance. A natural estimator of the sandwich covariance is the plug-in estimator $\hat{V}^g := \hat{\mathcal{I}}^{-1}\hat{\Sigma}\hat{\mathcal{I}}^{-1}$, which consistently estimates V^g. However, computing $\hat{V}^g$ requires the MAP $\hat{\theta}$, which is not typically computable directly from MCMC samples.

The reason that V^{IJ} might be expected to approximate V^g asymptotically can be seen by applying the delta method and the CLT results of Eq. 4 to the influence score ψ_n defined in Eq. 3. Specifically, for any particular n, we expect that

$$
\psi_n = N \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(\theta, \ell(x_n|\theta)\right) = \hat{\mathcal{I}}^{-1}\ell_{(1)}(x_n|\hat{\theta}) + O_p(N^{-1}). \tag{5}
$$

If Eq. 5 holds sufficiently uniformly in n (in a sense that we analyze precisely below), then

Eq. 5 implies the asymptotic equivalence of V^{IJ} and V^g:

$$\psi(x_n) \approx \hat{\mathcal{I}}^{-1}\ell_{(1)}(x_n|\hat{\theta}) \quad \Rightarrow \quad V^{\mathrm{IJ}} = \frac{1}{N}\sum_{n=1}^{N}\psi_n\psi_n^T - \bar{\psi}\,\bar{\psi}^T \approx \hat{\mathcal{I}}^{-1}\hat{\Sigma}\hat{\mathcal{I}}^{-1}, \tag{6}$$

and we know that $\hat{\mathcal{I}}^{-1}\hat{\Sigma}\hat{\mathcal{I}}^{-1} = \hat{V}^g \xrightarrow[N\to\infty]{prob} V^g$.

However, in general, the $O_p\left(N^{-1}\right)$ term in Eq. 5 depends on x_n, and the IJ covariance depends on *every* $\psi(x_n) : n \in [N]$, some of which may be pathologically behaved, especially under misspecification. A rigorous analysis of the IJ covariance requires understanding whether the approximation in Eq. 5 holds sufficiently strongly for us to apply the reasoning of Eq. 6. In particular, we must establish that we can safely sum the squares of the $O_p\left(N^{-1}\right)$ terms of Eq. 5. It is to this technical task that we now turn.

3.1 Consistency theorem

To prove the consistency of the IJ covariance, we assume the following conditions on the model and data generating process.

Assumption 1. *For $\delta > 0$, denote the δ–neighborhood of θ_∞ as $B_\delta := \{\theta : \|\theta - \theta_\infty\|_2 \le \delta\}$. Assume there exists a $\delta > 0$ such that:*

1. *The dimension of θ stays fixed as N changes.*
2. *The prior density $\pi(\theta)$ is non-zero and bounded for all $\theta \in \Omega_\theta$, and four times continuously differentiable in B_δ.*
3. *The prior is proper.*
4. *The functions $\mathscr{L}(\theta)$ and $\theta \mapsto \ell(x_*|\theta)$ are four times continuously differentiable in B_δ, and the interchange of partial differentiation with respect to θ and expectations with respect to $\mathbb{F}(\mathcal{X})$ is justified.*

5. *There exists a function $M(x_*)$ such that*

$$\sup_{\theta \in B_\delta} \left\| \ell_{(4)}(x_* | \theta) \right\|_2 \le M(x_*) \quad \textit{and} \quad \underset{\mathbb{F}(x_*)}{\mathbb{E}} \left[M(x_*)^2 \right] < \infty.$$

6. *The log likelihood has a strict optimum in the following sense:*

   - *A unique maximum $\theta_\infty = \mathrm{argmax}_{\theta \in \Omega_\theta} \mathscr{L}(\theta)$ exists.*
   - *The matrix $\mathcal{I}$ is positive definite.*
   - *There exists an $\varepsilon > 0$ such that, with probability approaching one as $N \to \infty$,*

$$\sup_{\theta : \Omega_\theta \setminus B_\delta} \left(\frac{1}{N} \sum_{n=1}^{N} \ell(x_n | \theta_\infty) - \frac{1}{N} \sum_{n=1}^{N} \ell(x_n | \theta) \right) > \varepsilon.$$

Under Assumption 1, a BCLT will apply in total variation distance by classical results (van der Vaart (2000) Chapter 10), but we require an expansion of posterior moments to analyze Eq. 6. Series expansions of posterior moments are studied by Kass, Tierney, and Kadane (1990) (under stricter differentiability conditions than ours), but with no control over the data dependence in the error when the expectation depends on the data.

We now state conditions on functions $\phi(\theta)$ and $\phi(\theta, x_n)$ sufficient to form a controllable series expansion of their posterior expectations.

Definition 2. A vector–valued function $\phi(\theta, x_*)$, possibly depending on a single datapoint, is "K-th order BCLT-okay" if

1. $\theta \mapsto \phi(\theta, x_*)$ is K times continuously differentiable $\mathbb{F}$–almost surely.
2. $\frac{1}{N} \sum_{n=1}^{N} \underset{\pi(\theta)}{\mathbb{E}} \left[\| \phi(\theta, x_n) \|_2^2 \right] = O_p(1)$.
3. For some $\delta > 0$, and for each $k = 1, \ldots, K$, $\sup_{\theta \in B_\delta} \frac{1}{N} \sum_{n=1}^{N} \left\| \phi_{(k)}(\theta, x_n) \right\|_2^2 = O_p(1)$.

When the function $\phi(\theta, x_n) = \phi(\theta)$ does not depend on the data (e.g., $\phi(\theta, x_n) = g(\theta)$), then Item 3 is vacuous, and Item 2 simply becomes $\underset{\pi(\theta)}{\mathbb{E}} \left[\| \phi(\theta) \|_2^2 \right] < \infty$. ◀

Theorem 1. *(See proof in Appendix D.) Let Assumption 1 hold, and assume that $\phi(\theta, x_n)$ is third-order BCLT-okay. Let $\hat{\mathcal{M}}$ denote the $D_\theta \times D_\theta \times D_\theta \times D_\theta$ array of fourth moments of the Gaussian distribution $\mathcal{N}(0, \hat{\mathcal{I}}^{-1})$. With $\mathbb{F}$-probability approaching one, for each n,*

$$\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}\left[\phi(\theta, x_n)\right] - \phi(\hat{\theta}, x_n) = N^{-1}\left(\frac{1}{2}\phi_{(2)}(\hat{\theta}, x_n)\hat{\mathcal{I}}^{-1} + \frac{1}{6}\phi_{(1)}(\hat{\theta}, x_n)\mathscr{L}_{(3)}(\hat{\theta})\hat{\mathcal{M}}\right) +$$
$$\tilde{O}\left(N^{-2}\right) 1_\phi \mathscr{R}^\phi(x_n), \text{ where } \frac{1}{N}\sum_{n=1}^{N}\mathscr{R}^\phi(x_n)^2 = O_p(1). \quad (7)$$

In Eq. 7, $\mathscr{R}^\phi(x_n)$ is a scalar, 1_ϕ is a vector of ones with the same dimension as $\phi(\theta, x_n)$, and derivatives with respect to θ are summed against the indices of the Gaussian arrays.[2]

To connect Theorem 1 to the IJ covariance, we first show that Assumption 1 is sufficient (indeed, stronger than necessary) to prove the (known) results of Eq. 4 for third–order BCLT-okay functions. For completeness, we state and prove these known results under our stated assumptions.

Assumption 2. *The functions $g(\theta)$, $\ell(x_*|\theta)$, and $g(\theta)\ell(x_*|\theta)$ are all third-order BCLT-okay.*

Proposition 1. *(See proof in Appendix C.2, as well as van der Vaart, 2000, Chapter 7) Let Assumptions 1 and 2 hold. Then $\hat{\theta} \xrightarrow[N\to\infty]{prob} \theta_\infty$, and*

$$\sqrt{N}\left(g(\hat{\theta}) - g(\theta_\infty)\right) \xrightarrow[N\to\infty]{dist} \mathcal{N}\left(0, V^g\right) \quad \text{where} \quad V^g = g_{(1)}(\theta_\infty)\mathcal{I}^{-1}\Sigma\mathcal{I}^{-1}g_{(1)}(\theta_\infty)^\intercal.$$

The next result shows that our target variance is V^g.

[2] Although a slight abuse of notation, these simple conventions avoid a tedious thicket of indicial sums. Concretely, we mean that

$$\phi_{(2)}(\hat{\theta}, x_n)\hat{\mathcal{I}}^{-1} = \sum_{a,b=1}^{D_\theta} \left.\frac{\partial^2\phi(\theta, x_n)}{\partial\theta_a\partial\theta_b}\right|_{\hat{\theta}} (\hat{\mathcal{I}}^{-1})_{ab}$$
$$\phi_{(1)}(\hat{\theta}, x_n)\mathscr{L}_{(3)}(\hat{\theta})\hat{\mathcal{M}} = \sum_{a,b,c,d=1}^{D_\theta} \left.\frac{\partial\phi(\theta, x_n)}{\partial\theta_a}\right|_{\hat{\theta}} \left.\frac{\partial^3\mathscr{L}(\theta)}{\partial\theta_b\partial\theta_c\partial\theta_d}\right|_{\hat{\theta}} \hat{\mathcal{M}}_{abcd}$$

each of which is a vector expression of the same length as $\phi(\theta, x_n)$.

Corollary 1. *(van der Vaart, 2000, Theorem 10.8) Under Assumptions 1 and 2,*

$$\sqrt{N}\left(\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[g(\theta)] - g(\theta_\infty)\right) \xrightarrow[N\to\infty]{dist} \mathcal{N}\left(0, V^g\right).$$

Proof. Under Assumptions 1 and 2, Lemma 4 in Appendix D yields that $g_{(1)}(\hat{\theta})$, $\hat{\mathcal{I}}$, and $\hat{\mathscr{L}}_{(3)}(\hat{\theta})$ are all $O_p(1)$. By Theorem 1 we thus have,

$$\sqrt{N}\left(\mathop{\mathbb{E}}_{\mathbb{P}(\theta|\mathcal{X})}[g(\theta)] - g(\theta_\infty)\right) = \sqrt{N}\left(g(\hat{\theta}) - g(\theta_\infty)\right) + O_p(N^{-1/2}),$$

The distributional limit then follows from Proposition 1 and Slutsky's theorem. $\square$

Finally, our main result, the consistency of V^{IJ}, follows from Theorem 1 and Corollary 1.

Theorem 2. *(See proof in Appendix E.) Under Assumptions 1 and 2 the infinitesimal jackknife variance estimate is consistent. Specifically,*

$$V^{\mathrm{IJ}} = g_{(1)}(\hat{\theta})\hat{\mathcal{I}}^{-1}\hat{\Sigma}\hat{\mathcal{I}}^{-1}g_{(1)}(\hat{\theta})^\intercal + \tilde{O}_p\left(N^{-1}\right) \xrightarrow[N\to\infty]{prob} V^g.$$

3.2 Discussion of the proof and results

Our proof approach closely follows classical proofs of the parametric BCLT, with the key difference that we keep track of the data dependence in the residuals (Chapter 10 of van der Vaart (2000), Theorem 1.4.2 of Ghosh and Ramamoorthi (2003), and Lehmann and Casella (2006)). A schematic of the proof setup is shown in Figure 2, in which three key regions, R_1, R_2 and R_3 are illustrated; these regions are formally defined in Definition 7 of Appendix D. As N increases, we choose the ball diameter δ small enough that three things happen:

- $\hat{\mathscr{L}}(\theta) \to \mathscr{L}(\theta)$ uniformly within B_δ (primarily thanks to Assumption 1 Item 5);
- $\hat{\mathscr{L}}(\theta)$ remains strictly bounded away from $\hat{\mathscr{L}}(\theta_\infty)$ outside B_δ (primarily thanks to Assumption 1 Item 6); and

- The posterior, which is proportional to $\exp(N\hat{\mathscr{L}}(\theta))$, concentrates in a region R_1 that shrinks at a rate slightly slower than $1/\sqrt{N}$.

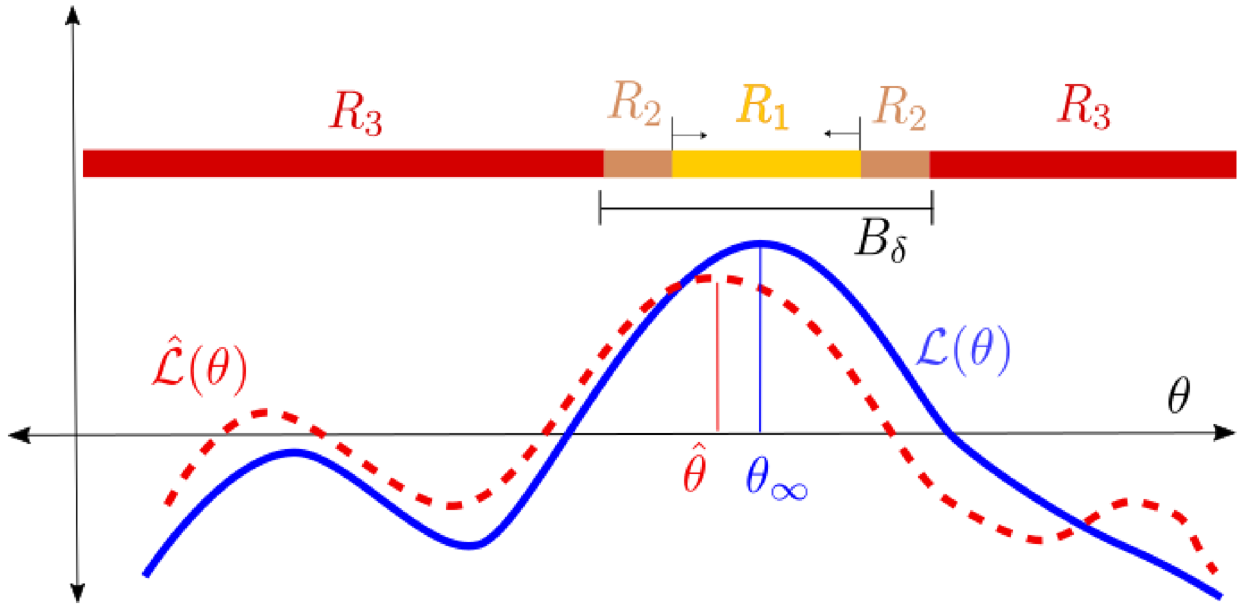

Figure 2: Visual illustration of the proof technique. As N increases, $\hat{\mathscr{L}}(\theta) \to \mathscr{L}(\theta)$ uniformly within B_δ, $\hat{\mathscr{L}}(\theta)$ remains bounded away from $\hat{\mathscr{L}}(\theta_\infty)$ outside of B_δ, and the region R_1 that matters for the posterior shrinks at a rate slightly slower than $1/\sqrt{N}$.

Taking as given that the posterior concentrates in the shrinking region R_1, we can fully characterize the posterior using Taylor series expansions within the region R_1. Our ability to form these Taylor series is guaranteed by the differentiability conditions of Assumption 1 Item 4 and Definition 2 Item 1. Our ability to control the terms in the Taylor series is guaranteed by uniform laws of large numbers (Chapter 19 of van der Vaart (2000)), which hold via Assumption 1 Item 5 and Definition 2 Item 3.

Finally, establishing that the posterior concentrates in R_1 requires separate treatment of the region R_2, which is compact but adjacent to the region of posterior concentration, and R_3, which encompasses the entire domain outside of B_δ. Thanks to uniform convergence within B_δ, the region R_2 can be dealt with via Taylor expansions and carefully choosing the shrinking rate of R_1. However, the region R_3 requires a suite of additional assumptions to guarantee that the posterior does not behave pathologically in the tails: namely, that

$\hat{\mathscr{L}}(\theta)$ remains strictly bounded away from $\hat{\mathscr{L}}(\theta_\infty)$ (Assumption 1 Item 6), that the prior is proper and bounded (Assumption 1 Items 2 and 3), and that average prior expectations of the function of interest exist (Definition 2 Item 2).

We note that most of our assumptions about the prior (boundedness, propriety, that certain prior expectations are bounded) serve only to establish that the unbounded region R_3 does not contribute meaningfully to the posterior. Inspection of the proof will reveal that weaker assumptions could also suffice, at the cost of increased technical or notational complexity. For example, since our analysis is asymptotic, one could assume only that the *posterior* satisfies Assumption 1 Items 2 and 3 and Definition 2 Item 2 with probability one after some finite number of observations. This posterior could then play the role of the prior in the rest of the proof without modification.

In Appendix A, we verify our assumptions and prove the consistency of the IJ covariance in the simple closed form example of inferring a univariate normal mean with a misspecified variance. This example, though simple, provides intuition as well as a template for more complex settings.

3.3 Data weights and the influence function

In this section, we briefly connect the IJ covariance to some related literature on local sensitivity, outlier detection, and the von Mises calculus. The connections are illuminated by the following "weighted posterior expectation"

$$\underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[g(\theta)] = \frac{\int_{\Omega_\theta} g(\theta) \exp\left(N \frac{1}{N}\sum_{n=1}^N w_n \ell(x_n|\theta)\right)\pi(\theta) d\theta}{\int_{\Omega_\theta} \exp\left(N \frac{1}{N}\sum_{n=1}^N w_n \ell(x_n|\theta)\right)\pi(\theta) d\theta}, \tag{8}$$

where $\mathcal{W} = (w_1, \ldots, w_N) \in \mathbb{R}^N$ is a length–N vector of scalar–valued weights, one for each datapoint. Comparing Eq. 1 with Eq. 8, we see that $\underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[g(\theta)] = \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$ when $\mathcal{W} = 1_N$, where 1_N is the length–N vector containing all ones. When we can apply the dominated convergence theorem to exchange posterior integration and differentiation (see, e.g., Theorem 1 of Giordano, Broderick, and Jordan (2018)), it follows that

$$\left.\frac{\partial \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[g(\theta)]}{\partial w_n}\right|_{\mathcal{W}=1_N} = \underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}\left(g(\theta), \ell(x_n|\theta)\right) = \frac{1}{N}\psi_n. \tag{9}$$

Our "influence scores" are thus the values of the "empirical influence function" of the statistical functional $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[g(\theta)]$, and our V^{IJ} the classical IJ covariance estimator (see, e.g., von Mises (1947), Reeds (1976), Efron (1982), and Hampel et al. (2011), or Serfling (1980) for a textbook treatment). Equivalently, the empirical influence function can be understood as an instance of "local sensitivity analysis" to perturbations of data weights (Gustafson, 1996; Giordano, Broderick, and Jordan, 2018), which motivates application of the empirical influence function to a wider variety of problems than variance estimation. The related literature is too large to survey here, but some examples are cross–validation (Koh and Liang, 2017; Giordano, Stephenson, et al., 2019), identification of outliers (Pregibon, 1981; McCulloch, 1989; Belsley, Kuh, and Welsch, 2005), identification of influential subsets (Giordano, Meager, and Broderick, 2026a; Giordano, Meager, and Broderick, 2026b), and interpretation of Bayesian posteriors (Thomas, MacEachern, and Peruggia, 2018; Iba, 2025). The present work rigorously connects Bayesian posterior expectations to this large literature, and we anticipate applications beyond variance estimation.

However, the interpretation of ψ_n as a local approximation, together with the form of Eq. 8, provides a cautionary warning in the Bayesian context. Though our experiments of Section 4 motivate the IJ covariance estimator as a way to avoid pathologies associated with MAP estimation, the most common use of MCMC is to integrate out high–dimensional posterior parameters that do not obey a BCLT. The following simple argument suggests that the IJ covariance may not work in such a case, even for parameters that do obey a BCLT marginally.

Suppose, for example, that θ is the concatenation of two sets of parameters: a finite–dimensional set of "global" parameters γ, and a high–dimensional set of "local" parameters, λ. A common such example is where γ are "fixed effects" and λ "random effects" in a panel data setting, and the full parameter set is the concatenation $\theta = (\gamma, \lambda)$ (see, e.g. Davidson and MacKinnon (2004, Chapter 7), though in a Bayesian context the distinction between fixed and random effects is not always meaningful (Gelman and Hill, 2006, Chapter 11)). Suppose that $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}(\gamma) = \tilde{O}_p(N^{-1})$ but $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathrm{Cov}}(\lambda)$ remains bounded away from zero. Take $g(\theta) = \gamma$. Differentiating Eq. 9 again with respect to w_n and applying the tower property and delta method gives:

$$\left.\frac{\partial^2 \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\gamma]}{\partial w_n^2}\right|_{\mathcal{W}=1_N} = \underset{\mathbb{P}(\gamma|\mathcal{X})}{\mathrm{Cov}}\left(\gamma, \underset{\mathbb{P}(\lambda|\gamma,\mathcal{X})}{\mathrm{Cov}}(\ell_n(\gamma,\lambda))\right) + \tilde{O}_p\left(N^{-2}\right) \tag{10}$$

where the $\tilde{O}_p(N^{-2})$ term assumes that third–order posterior cumulants of $\mathbb{P}(\gamma|\mathcal{X})$ are $\tilde{O}_p(N^{-2})$, as they would typically be for a posterior which concentrates at root $N^{-1/2}$. By the tower property, the first derivative is also a posterior covariance with respect to $\mathbb{P}(\gamma|\mathcal{X})$:

$$\left.\frac{\partial \underset{\mathbb{P}(\theta|\mathcal{X};\mathcal{W})}{\mathbb{E}}[\gamma]}{\partial w_n}\right|_{\mathcal{W}=1_N} = \underset{\mathbb{P}(\gamma|\mathcal{X})}{\mathrm{Cov}}\left(\gamma, \underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathbb{E}}[\ell(x_n|\gamma,\lambda)]\right). \tag{11}$$

See Appendix F for details on the derivation of Eqs. 10 and 11.

If the function $\gamma \mapsto \underset{\mathbb{P}(\lambda|\gamma,\mathcal{X})}{\mathrm{Cov}}(\ell_n(\gamma,\lambda))$ does not vanish as $N \to \infty$, it follows from Eqs. 10 and 11 that both the first and second derivatives are posterior covariances of non–vanishing functions of γ, and so of the same order as $N \to \infty$. Since the second derivative is of the same order as the first derivative, the empirical influence function's linear approximation to provide a poor approximation even to the effect of leaving out a single datapoint.

Note that this counterexample does not contradict our results from Theorem 2: if a BCLT applies to λ as well as to γ, then $\underset{\mathbb{P}(\lambda|\mathcal{X},\gamma)}{\mathrm{Cov}}(\ell(x_n|\gamma,\lambda)) = \tilde{O}_p(N^{-1})$ inside the covariance of Eq. 10, and the second derivative is $\tilde{O}_p\left(N^{-2}\right)$, as suggested by Theorem 1. This intuition is rigorously developed and supported using von Mises expansions in Giordano and Broderick (2024), which at the time of writing is still in progress. In light of this example, and of the analysis of Giordano and Broderick (2024) for the high–dimensional regime, we recommend caution when employing the IJ covariance estimator — or any method depending on the empirical influence function — in settings where a BCLT may not apply to the full posterior $\mathbb{P}(\theta|\mathcal{X})$.

3.4 Related Work

Several other papers have studied the estimation of frequentist variability of posterior expectations. Efron (2015) provides results similar to ours, but is limited to the special case of exponential families. The frequentist sampling variability of Bayesian posterior expectations under misspecification is studied in Kleijn and van der Vaart (2012) from a theoretical point of view, motivating the form of the sandwich covariance. Huggins and Miller (2024) derive a BCLT for the "BayesBag" posterior which combines both posterior and bootstrap uncertainty; we conjecture that a minor modification of their proof would prove the consistency of the bootstrap alone as an estimator of the limiting frequentist variance of a posterior mean. See also Huggins and Miller (2023) for a closely related BCLT result in the context of model selection, as well as the same authors' earlier work on arxiv, Huggins and Miller (2020). Although the IJ covariance does not provide bootstrap samples for use in BayesBag posteriors directly, the IJ covariance could be used to estimate the bootstrap covariance term that appears in the law of total covariance decomposition of the BayesBag covariance (see the final, unnumbered equation in Section 2.3 of Huggins and Miller (2024)). Both Vehtari et al. (2024) and D. Lee, Carroll, and Sinha (2017) consider importance sampling estimates to the effect of dropping datapoints, with D. Lee, Carroll, and Sinha (2017) explicitly considering estimation of frequentist variability of the posterior under an assumption of correct specification. However, importance sampling estimates of frequentist variability can be expected to have very low sampling efficiency even in moderate dimensional problems, since frequentist re–sampling is expected to produce changes of the posterior mean on the same order as the posterior standard deviation (see Example 9.2 of Owen (2013)). Furthermore, misspecification can result in frequentist sampling variances that are different than the posterior variance, which can further degrade importance sampling efficiency as a function of dimension (see Example 9.3 of Owen (2013)).

Our key result, Theorem 1, is distinguished from previous BCLT results primarily by tracking the data dependence in the residual of the series expansion, though we also differ from some classical work by admitting misspecification. Our method of proof hews closely to the classical BCLT proofs found in Chapter 10 of van der Vaart (2000), Theorem 1.4.2 of Ghosh and Ramamoorthi (2003), and Lehmann and Casella (2006), all of which study the asymptotic regime for a single posterior quantity under the assumption of correct specification. A series expansion for posterior expectations similar to Theorem 1 is established by Kass, Tierney, and Kadane (1990), under stricter differentiability conditions than ours, and only for a single posterior quantity of interest, which is (implicitly) assumed not to depend on the dataset. A number of recent papers have, like us, studied the asymptotic behavior of finite–dimensional Bayesian posteriors under misspecification, though without explicitly controlling data dependence (Kleijn and van der Vaart, 2006; Miller, 2021; Kasprzak, Giordano, and Broderick, 2025).

Our work is an application of the classical IJ estimator, which has been extensively studied in the frequentist literature, but which to the best of the authors' knowledge has not been applied to Bayesian posterior expectations (see, e.g., Jaeckel (1972), Efron (1982), Shao and Tu (2012), Efron (2014), Wager, Hastie, and Efron (2014), Peng, Mentch, and Stefanski (2024), Ghosal, Zhou, and Hooker (2026)). As we discuss more explicitly in Section 3.3, the IJ covariance can be viewed as the first term in a von Mises expansion of the Bayesian posterior (von Mises (1947), Reeds (1976), Serfling (1980)), though we prove consistency directly rather than via a functional series expansion. Giordano and Broderick (2024) prove consistency of the IJ for posterior expectations using von Mises expansions directly under stronger theoretical assumptions, using results from the present paper.

As we discuss in the introduction, IJ estimation can be thought of as a particular measure of local data robustness — specifically, robustness to random resampling of the data. We make this connection even more explicit in Section 3.3. Viewing IJ estimators as local data robustness connects our work to a large literature on "local case influence," which refers to a broad class of techniques that using derivatives to approximate the effect of a datapoint or datapoints on a statistic of interest. Local case influence is an idea with a long history in frequentist and Bayesian statistics. The literature is too vast to review systematically here, but some notable references include Belsley, Kuh, and Welsch (2005), Cook (1977), Cook and Weisberg (1982), Cook (1986), Kempthorne (1986), Kass, Tierney, and Kadane (1989), Carlin and Polson (1991), Zhu, Ibrahim, S. Lee, et al. (2007), Koh and Liang (2017), and Thomas, MacEachern, and Peruggia (2018). The relationship between posterior covariances and derivatives has been noted numerous times in the Bayesian robustness literature (e.g. Diaconis and Freedman (1986), Gustafson (1996), Basu, Jammalamadaka, and Liu (1996), Efron (2015), Giordano, Broderick, and Jordan (2018)). As we note in Section 3, consistency of the IJ requires a kind of uniform guarantee over a large number of local approximations, but most of the local case influence work focuses on a datapoint or a small set of datapoints. Notable exceptions include Giordano, Stephenson, et al. (2019) in the frequentist literature and Zhu, Ibrahim, Cho, et al. (2012) in the Bayesian literature, though we note that Theorem 2 of Zhu, Ibrahim, Cho, et al. (2012) is stated in terms of a fixed set of dropped datapoints.

Following the posting of the present work on the arXiv, a number of interesting subsequent papers have proposed extensions and variants and applications of the Bayesian IJ. Nguyen et al. (2024) applies the IJ to the adversarial data–dropping work of Giordano, Meager, and Broderick (2026a) and Giordano, Meager, and Broderick (2026b) in MCMC settings. Iba (2025) connects the IJ covariance to kernel methods, and Iba and Yano (2023) apply ideas closely related to the Bayesian IJ covariance to information criteria and model selection. Two papers, F. Ji, J. Lee, and Rabe-Hesketh (2025) and Luo and Feng Ji (2026), have systematically evaluated the Bayesian IJ covariance for use in the social sciences in the presence of model misspecification. We hope that our approach continues to motivate and inform connections between the frequentist and Bayesian case influence literature.

4 Experiments

We present two experimental analyses of the IJ covariance. The first is a simulated random effects model under misspecification. We designed the simulation to have a degenerate MAP approximation but well-behaved MCMC posterior expectation, and show that the IJ covariance is able to accurately estimate the true sampling variance over new draws of the data. The second experiment is an application of the IJ covariance to a large number of linear and generalized linear models and datasets taken from Gelman and Hill (2006). On these real-world datasets a simulated ground truth is unavailable, so we compare the IJ covariances to the much more computationally expensive bootstrap estimated variance.

Additional experimental results can be found in Appendix H, including a number of closed–form examples, as well as an additional application to a real-world "multilevel regression and post–stratification" analysis from the surveys literature (Park, Gelman, and Bafumi, 2004).

4.1 A degenerate joint MAP approximation with simulated data

As we describe in Section 2.2, the IJ covariance is particularly well-motivated in settings when the MAP $\hat{\theta}$ is either unavailable or unreliable, and one must turn to MCMC to integrate out a large number of nuisance parameters. In this section we present a simple simulated example of such a case.

Consider the following simple random effects model with responses. For the data, we assume we have responses $y_n \in \mathbb{R}$, regressors $r_n \in \mathbb{R}$, and group indicator vectors $z_n \in \{0,1\}^K$, for $n = 1, \ldots, N$ and some fixed number of groups K. We assume that all three are conditionally IID, so $x_n = (y_n, r_n, z_n)$, and draw according to the model

$$\varepsilon_n \stackrel{indep}{\sim} \mathcal{N}(\cdot|0,1) \qquad r_n \stackrel{indep}{\sim} \mathcal{N}(\cdot|0,1) \qquad y_n = r_n + r_n^2 \varepsilon_n. \tag{12}$$

In our experiments we used $K = 100$ and $N = 10{,}000$.

Suppose we now form a misspecified model of this data. Define the regression parameter $\beta \in \mathbb{R}$, random effect vector $\mu \in \mathbb{R}^K$, and variances σ_y^2 and σ_μ^2. We will take the priors on μ_k to be independent given σ_μ, and the priors on β, σ_y and σ_μ to be diffuse; we will shortly be using the default diffuse priors provided by `rstanarm`. Then we can model

$$y_n | z_n, r_n, \beta, \mu, \sigma_y^2 \stackrel{indep}{\sim} \mathcal{N}(\cdot|\beta r_n + z_n^T \mu, \sigma_y^2) \quad \text{and} \quad \mu_k | \sigma_\mu \stackrel{indep}{\sim} \mathcal{N}(\cdot|0, \sigma_\mu^2). \tag{13}$$

The model in Eq. 13 is misspecified because it does not model the heteroskedasticity induced by $r_n^2 \varepsilon_n$ in Eq. 12. Note also that the group indicators z_n have no effect on the real data generating process, and so estimates of very small σ_μ should be supported by the data — and the log likelihood should be degenerate at the point σ_μ, as we describe in detail below.

How good is the MAP estimate for this model? The answer depends on which variables are marginalized in the definition of θ. Suppose, for example, we form the "global" MAP, optimizing over all the parameters, including the random effects:

$$\hat{\beta}^{\mathrm{glob}}, \hat{\sigma}_y^{\mathrm{glob}}, \hat{\sigma}_\mu^{\mathrm{glob}}, \hat{\mu}^{\mathrm{glob}} := \operatorname{argmax} \ell(\mathcal{X}|\beta, \sigma_y, \sigma_\mu, \mu) + \log \pi(\mu|\sigma_\mu) + \log \pi(\beta, \sigma_y, \sigma_\mu).$$

Optimizing over all parameters including the random effects is prone to under-estimate the variance σ_y, particularly when K is large relative to N. A simplified version of this phenomenon is the known as the Neyman-Scott paradox (Freedman, 2005). For this reason, one does not typically optimize over μ for such a model. Rather, one optimizes the marginal posterior (or an approximation thereof):

$$\begin{aligned}\hat{\beta}, \hat{\sigma}_y, \hat{\sigma}_\mu &:= \operatorname{argmax} \log \int p(\mathcal{X}|\beta, \sigma_y, \sigma_\mu, \mu)\pi(\mu|\sigma_\mu) d\mu + \log \pi(\beta, \sigma_y, \sigma_\mu) \\ &= \operatorname{argmax} \sum_{k=1}^{K} \ell(\mathcal{X}_k|\beta, \sigma_y, \sigma_\mu) + \log \pi(\beta, \sigma_y, \sigma_\mu), \end{aligned} \tag{14}$$

where $\mathcal{X}_k$ denotes $\{x_n : z_{nk} = 1\}$, the observations in group k. For Gaussian models such as this one, the needed integral in Eq. 14 can be computed in closed form. In the general case, there exist non-MCMC methods such as marginal maximum likelihood estimation (MMLE) use the Laplace approximation to approximate the integral (Bates et al., 2015). However, it is possible to generate data $\mathcal{X}$ so that even the MAP of Eq. 14 estimates $\hat{\sigma}_\mu = 0$ for sufficiently non-informative priors (Chung et al., 2012; Eager and Roy, 2017).

In this case, we have designed the model and data generating process so that the MMLE package `lme4` results in a numerically singular fit ($\hat{\sigma}_\mu = 0$) approximately half the time. When $\hat{\sigma}_\mu = 0$, the second derivative of the log posterior is degenerate at the optimum, and classical methods for approximating frequentist variances cannot be directly applied. In a random effects model for which even MMLE methods fail, MCMC is a natural recourse. If an estimate of the frequentist variability of the posterior mean is required, the IJ covariance is then a natural choice.

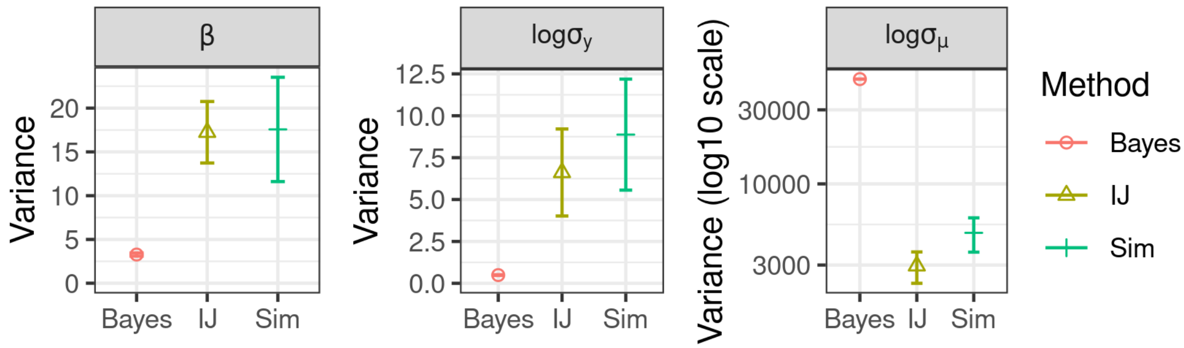

Figure 3: The difference from the simulation, IJ, and Bayesian covariances. Standard errors include variation due to both Monte Carlo and, in the case of the IJ and simulation, frequentist error.

We simulated a single dataset for which the `lme4` fit happened to be singular, and computed 10,000 MCMC draws from the model of Eq. 13 using `rstanarm`. With these samples, we computed the Bayesian posterior covariance and the IJ covariance. We then simulated 100 different datasets, ran the same MCMC procedure for each, computed the posterior mean for each simulation, and computed the covariance of the posterior means over the simulations. Ideally, we expect the IJ covariance to provide a good approximation to the simulation-based covariance estimate.

The results are shown in Figure 3. As shown in the left plot, the posterior variances of $\log \sigma_y$ and of β are dramatically less than the frequentist variances of $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\log \sigma_y]$ and $\underset{\mathbb{P}(\theta|\mathcal{X})}{\mathbb{E}}[\beta]$. The posterior of the log random effect variance, $\log \sigma_\mu$, is extremely non-normal, and the IJ may not expected to perform well, theoretically. Nevertheless, we show that the IJ covariance is correct despite the posterior variance being extremely inflated.

4.2 Applied regression modeling examples

We will now evaluate the performance of the IJ covariance on a number of real-world models and datasets taken from Gelman and Hill (2006). We limited our analysis to models for which data was available in the Stan examples repository (Stan Team, 2017), and for which `rstanarm` (Goodrich et al., 2020) was able to run in a reasonable amount of time. Since these are real-world datasets, the ground truth sampling distribution is not available from simulations, so, as an imperfect proxy, we compare the IJ covariance to the (computationally expensive) bootstrap variance instead.

All models we considered from Gelman and Hill (2006) were instances of generalized linear models, including both ordinary and logistic regression as well as models with only fixed effects and models that include random effects. Model details, including the number of estimated quantites of each type, are given in Table 2 of Appendix G.1. For parameters of interest, $g(\theta)$, we took all fixed-effect regression parameters, the log of the residual variance (when applicable), and the log of the random effect variance (when applicable). We did not examine random effects, nor the random effect covariances for the few models with multiple random effects, out of concern that a BCLT might not be expected to hold for these parameters. Below, when we refer to a "parameter," we refer to a scalar parameter of interest in a particular model for a particular dataset; there were a total of 768 such parameters.

For each model and dataset, we fix a number of MCMC samples, $M = 2{,}000$ to draw for each posterior sample, and a number of bootstraps samples to draw, $B = 200$. The total computation time summed across all models for the initial MCMC and IJ computation was 57.74 minutes, and the total bootstrap compute time summed across all models and bootstrap samples was 266.83 hours.[3] For each parameter of interest, we use a single run of MCMC to compute the IJ covariance estimate, $\hat{V}^{\mathrm{IJ}}$. Using Algorithm 2, we draw B different bootstrap samples, compute M MCMC samples for each, and finally compute the bootstrap variance estimate, $\hat{V}^{\mathrm{Boot}}$.

All together, this procedure produces comparisons between 768 distinct covariance

[3]Naively, one would expect the total bootstrap compute time to be roughly $B = 200$ times the compute time of the single MCMC run, but the total bootstrap time was in fact longer than this. We conjecture that the bootstrap time scaled super-linearly due to the fact that some bootstrap samples led to posteriors that were less well-conditioned than the original datasets, and so incurred more rejections during the MCMC procedure.

estimates from 56 different models using 14 distinct datasets. The median number of observations amongst the models was 240, and the range was from 40 to 3,020 exchangeable observations. Since we fit our models in `rstanarm`, we were able to use the function `rstanarm::log_lik` to compute $\ell(x_n|\theta)$, so $\hat{V}^{\mathrm{IJ}}$ required essentially no additional coding. We used the default priors provided by `rstanarm`.

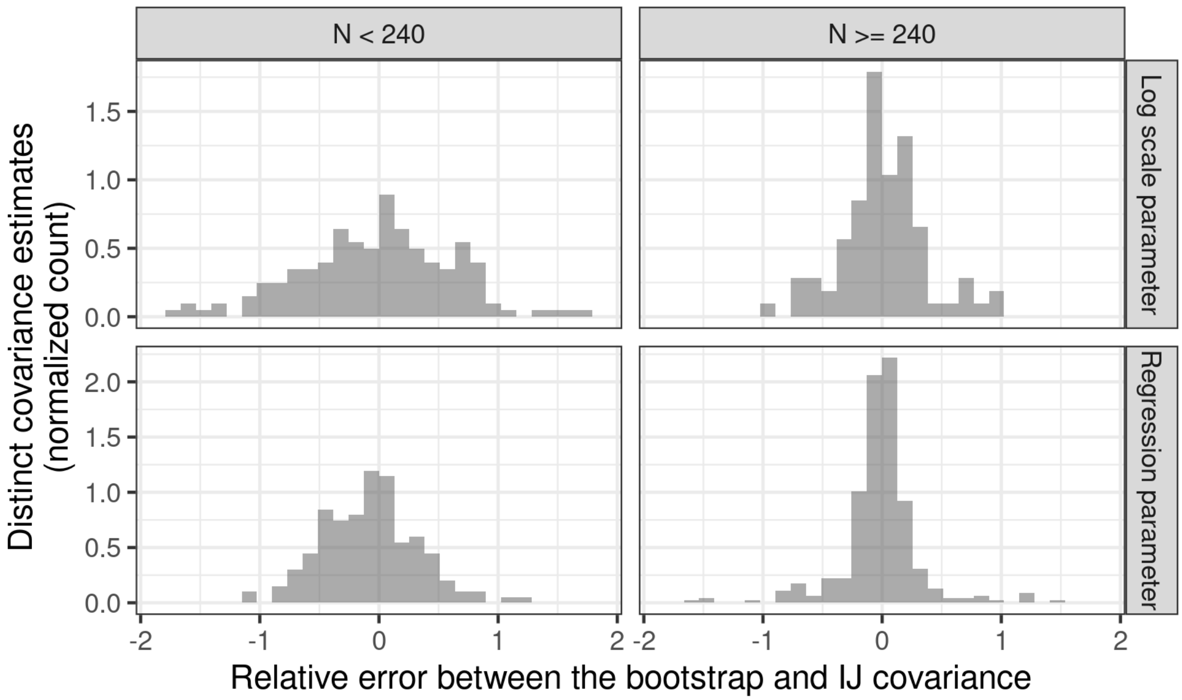

Figure 4: The distribution of the relative errors Δ^{IJ}. The "Log scale parameter" panel includes all variances or covariances that involve at least one log scale parameter.

With 768 covariances to report, we must summarize our results into a few model–agnostic high-level accuracy summaries. To this end, for each parameter j in each model i, we computed the following measure of both the "practical significance" and "statistical significance" of the IJ covariance covariance as a proxy for the bootstrap variance:

$$\Delta_{ij}^{\mathrm{IJ}} := \frac{\hat{V}_{ij}^{\mathrm{IJ}} - \hat{V}_{ij}^{\mathrm{Boot}}}{\left|\hat{V}_{ij}^{\mathrm{Boot}}\right| + \Xi_{ij}^{\mathrm{boot}}} \quad \text{and} \quad Z_{ij} := \frac{\hat{V}_{ij}^{\mathrm{IJ}} - \hat{V}_{ij}^{\mathrm{Boot}}}{\sqrt{(\Xi_{ij}^{\mathrm{IJ}})^2 + (\Xi_{ij}^{\mathrm{boot}})^2}}. \tag{15}$$

In Eq. 15, the quantities Ξ^{IJ} and Ξ^{boot} are measures of the Monte Carlo error in the

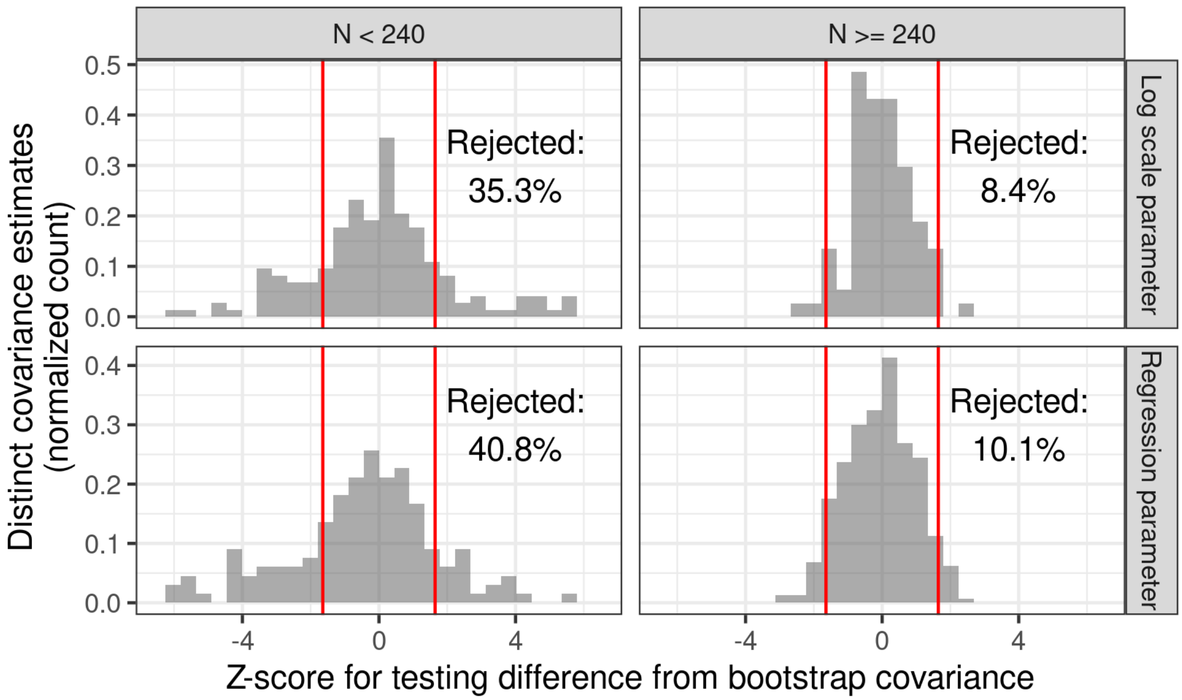

Figure 5: The distribution of the z-statistics Z. Red lines indicate the boundaries of a two-sided normal test for significance with level 0.1, and "Rejected" counts the number of covariances in both sides of the rejection region. The "Log scale parameter" panel includes all variances or covariances that involve at least one log scale parameter.

corresponding covariance estimates (see Appendix G.0.1 for details).

The metric $\Delta_{ij}^{\mathrm{IJ}}$ is a (noisy) measure of the proportional difference between $\hat{V}_{ij}^{\mathrm{IJ}}$ and $\hat{V}_{ij}^{\mathrm{Boot}}$. The term Ξ_{ij}^{boot} is included to retain a meaningful notion of scale when $\hat{V}_{ij}^{\mathrm{Boot}}$ is nearly zero due to chance alone. The metric Z_{ij} is the z-statistic which we would compute to test equality between $\hat{V}_{ij}^{\mathrm{IJ}}$ and V_{ij}^{Boot} using our estimates of the sampling error. Despite this interpretation, the z-statistics for different parameters within a given model are based on the same MCMC samples and so are not independent, and we should be cautious in interpreting the collection of Z_{ij} as the basis of any kind of formal hypothesis test. Rather, we recommend thinking of Z_{ij} as heuristics whose marginal distributions summarize the error $\hat{V}^{\mathrm{IJ}} - V^{\mathrm{Boot}}$ relative to Monte Carlo error.

The distributions of $\Delta_{ij}^{\mathrm{IJ}}$ and Z_{ij} across models, datasets, and parameters are shown in 4 and 5 respectively. For models with more data, the IJ provides a reasonable approximation to the much more computationally expensive bootstrap, with roughly half of the IJ covariance estimates falling well within 50% of the bootstrap covariance estimates, and rejecting the hypothesis test of equivalence at roughly the nominal rate. For models with less data, IJ estimates do not appear to systematically deviate from the bootstrap estimates, but are more variable. Recall that we expect, theoretically, only that the IJ and bootstrap variances coincide as $N \to \infty$.

The distribution of Z_{ij} suggests that the apparent differences between the IJ and bootstrap can be accounted for by Monte Carlo error alone in the large models but not necessarily the small ones. Recalling that both the bootstrap and the IJ covariances are distinct approximations of the same quantity which are close to one another only asymptotically, we conjecture that the needed asymptotics have not yet kicked in for models with fewer datapoints.

5 Conclusion

We define and analyze the IJ covariance estimate of the frequentist variance of Bayesian posterior expectations. The IJ covariance is computationally appealing, since it can be computed easily from only a single set of MCMC draws, in contrast to the bootstrap, which, in general, requires many distinct MCMC chains. In finite dimensional problems obeying a central limit theorem, we show that the IJ covariance is consistent, since it consistently estimates the sandwich covariance. Our work rigorously connects Bayesian posterior inference with a large and growing literature on applications of the empirical influence function.

We caution that our proofs are asymptotic, based on finite-dimensional posteriors, and unfortunately come with no accompanying guarantees for any particular dataset. Based on our experience, we are confident recommending the IJ covariance as a rough but computationally efficient check for worryingly high levels of frequentist sampling variance. However, in applications where precise frequentist coverage is crucial, we recommend supplementing the IJ covariance with more computationally expensive procedures such as the bootstrap or conformal intervals (Huggins and Miller, 2023; Fong and Holmes, 2021). Furthermore, as we discuss in Section 3.3 above, a simple argument suggests that the IJ covariance will be inaccurate in high-dimensional posteriors, and the IJ should be used with extreme caution in such cases. Forming diagnostics to alert the user when IJ covariance may be inaccurate is an exciting and important avenue for future work.

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7 Data Availability Statement

A complete copy of the code and data needed to reproduce our experiments can be found in the github repository `https://github.com/rgiordan/BayesIJPaper` and in the paper's supplementary material.

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